

Frank Wolfe and RSD Algorithm



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Large Scale Optimization

Assignment 5

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Abstract

The objective of this assignment is to implement and compare two algorithms for the traffic equilibrium problem: the Frank-Wolfe algorithm and the Restricted Simplicial Decomposition algorithm. The network data, including nodes, links and base link costs, are taken from the previous assignments. In this work, congestion is introduced through linear volume-delay functions on all links.

The implementation is carried out in AMPL. The Frank-Wolfe algorithm is used as a classical projection-free method for the Beckmann formulation of the user equilibrium problem. The Restricted Simplicial Decomposition method is also implemented, using a restricted set of active vertices and solving a master problem over their convex hull.

The numerical results show that both methods approach the same equilibrium solution. Frank-Wolfe converges in 101 iterations, while the RSD method converges in 7 iterations. The solution is also compared with the direct non-linear solution obtained with AMPL. Finally, several origin-destination paths are analysed in terms of their final travel costs in order to check the Wardrop equilibrium condition.

1 Introduction

This assignment focuses on the implementation of two methods for solving a traffic equilibrium problem: the Frank-Wolfe algorithm and the Restricted Simplicial Decomposition algorithm. The problem is defined on the same network used in the previous assignments, with the same set of nodes, directed links, origins and destinations.

The main difference with respect to the previous network flow formulations is that link costs are no longer constant. Instead, the cost of each link depends on the total flow using that link. Therefore, the problem becomes nonlinear. The volume-delay functions are assumed to be linear and of the form

$$s_{ij}(v_{ij}) = c_{ij} + dv_{ij},$$

where c_{ij} is the base cost of link (i, j) , v_{ij} is the total flow on that link and d is a congestion parameter chosen in the interval required by the assignment.

The purpose of the implementation is not only to obtain the equilibrium solution, but also to compare the computational behaviour of Frank-Wolfe and RSD. For this reason, the origin-destination demands were increased in order to obtain a sufficiently congested instance. This is important because the performance of RSD is not very meaningful on a low-congestion network.

The implementation files used in this work are `MinCM2_completed.mod`, `net_previous_assignments.` and `MinCM2_FW_RSD_v4.run`. The numerical outputs are stored in `FW_results.txt` and `RSD_results.txt`.

2 Algorithm Formulation

2.1 Network Equilibrium Model

Let N be the set of nodes and let A be the set of directed links. The set of origin-destination pairs is denoted by OD . For every pair $(o, d) \in OD$, a demand g_{od} must be sent from origin o to destination d .

Let x_{ij}^{od} be the flow associated with origin-destination pair (o, d) on link (i, j) . The total flow on link (i, j) is

$$v_{ij} = \sum_{(o,d) \in OD} x_{ij}^{od}.$$

The flow conservation constraints are

$$\sum_{j:(i,j) \in A} x_{ij}^{od} - \sum_{j:(j,i) \in A} x_{ji}^{od} = b_i^{od}, \quad \forall i \in N, \forall (o, d) \in OD,$$

where

$$b_i^{od} = \begin{cases} g_{od}, & i = o, \\ -g_{od}, & i = d, \\ 0, & \text{otherwise.} \end{cases}$$

All flow variables are non-negative:

$$x_{ij}^{od} \geq 0.$$

The link cost functions are linear:

$$s_{ij}(v_{ij}) = c_{ij} + dv_{ij}.$$

The base cost c_{ij} is computed from the node coordinates, as in the previous assignments:

$$c_{ij} = 100 + (x_i - x_j)^2 + (2y_i - y_j)^2.$$

The user equilibrium problem can be written using the Beckmann transformation:

$$\min_v \sum_{(i,j) \in A} \int_0^{v_{ij}} s_{ij}(u) du.$$

Since

$$s_{ij}(u) = c_{ij} + du,$$

we obtain

$$\int_0^{v_{ij}} (c_{ij} + du) du = c_{ij}v_{ij} + \frac{d}{2}v_{ij}^2.$$

Therefore, the nonlinear objective function is

$$\min_x \sum_{(i,j) \in A} \left(c_{ij}v_{ij} + \frac{d}{2}v_{ij}^2 \right),$$

subject to the flow conservation constraints and non-negativity constraints.

2.2 Frank-Wolfe Algorithm

The Frank-Wolfe algorithm solves the nonlinear equilibrium problem by repeatedly solving linearized subproblems. At iteration k , the current flow is denoted by v^k . The gradient of the Beckmann objective is the vector of current link travel costs:

$$\nabla f(v^k)_{ij} = c_{ij} + dv_{ij}^k.$$

The linearized subproblem is

$$\min_{y \in X} \nabla f(v^k)^T y,$$

where X is the feasible set defined by the network flow conservation constraints. In traffic assignment terms, this corresponds to assigning all origin-destination demands to shortest paths computed with the current link costs.

Once the auxiliary flow y^k is obtained, the next iterate is computed as

$$v^{k+1} = v^k + \alpha_k(y^k - v^k),$$

where $\alpha_k \in [0, 1]$ is found by line search.

The relative gap is used as stopping criterion:

$$\text{gap} = \frac{UB - LB}{UB}.$$

The algorithm stops when the relative gap is below 0.0005 or when the maximum number of iterations is reached.

2.3 Restricted Simplicial Decomposition Algorithm

The Restricted Simplicial Decomposition algorithm is also based on linearized subproblems, but instead of updating only along the segment between the current solution and one new vertex, it keeps a restricted set of active vertices.

At each iteration, a new vertex is obtained by solving the linearized subproblem

$$\min_{y \in X} \nabla f(v^k)^T y.$$

This new vertex is added to the active set. If the maximum number of active vertices is exceeded, the vertex with the smallest barycentric coordinate is removed. In the implementation, the active set size is controlled by the parameter ρ .

The master problem is then solved over the convex hull of the active vertices:

$$\min_{\alpha} f \left(\sum_{r \in W} \alpha_r v^r \right),$$

subject to

$$\sum_{r \in W} \alpha_r = 1, \quad \alpha_r \geq 0.$$

The new iterate is therefore

$$v^{k+1} = \sum_{r \in W} \alpha_r v^r.$$

The barycentric coordinates α_r obtained in the master problem are reported at every iteration, together with the number of active vertices.

3 Results of the Implementations

3.1 Experimental Setup

The network contains 24 nodes and the directed links used in the previous assignments. The origins are nodes 3 and 11, and the destination nodes are

$$5, 8, 12, 13, 14, 17, 18, 19, 21, 23.$$

The origin-destination demands were increased in order to obtain a congested instance. The final run uses the data file `net_previous_assignments.dat` and the implementation file `MinCM2_FW_RSD_v4.run`.

The stopping tolerance for both algorithms is

$$\varepsilon = 0.0005.$$

The final results are compared with the direct nonlinear solution obtained by solving the Beckmann formulation directly with AMPL.

3.2 Frank-Wolfe Results

The Frank-Wolfe algorithm converged after 101 iterations. The final objective value was

$$z_{FW} = 24\,869\,762.38,$$

with a final relative gap equal to

$$gap_{FW} = 0.000463.$$

The behaviour observed is consistent with the theoretical properties of the Frank-Wolfe algorithm. During the first iterations the objective function decreases rapidly, but convergence becomes significantly slower near the optimum. In order to obtain a sufficiently challenging instance, the origin-destination demands were increased, generating a highly congested network.

As a consequence, more than one hundred iterations were required before the stopping tolerance was reached, which allows a meaningful comparison with the Restricted Simplicial Decomposition algorithm.

3.3 Restricted Simplicial Decomposition Results

The Restricted Simplicial Decomposition algorithm with parameter

$$\rho = 4$$

converged after only 7 iterations.

The final objective value obtained was

$$z_{RSD} = 24\,862\,823.17,$$

with a final relative gap

$$gap_{RSD} = 0.000356.$$

The final active barycentric coordinates were

$$\alpha_4 = 0.2831, \quad \alpha_5 = 0.2556, \quad \alpha_7 = 0.1819, \quad \alpha_8 = 0.1843.$$

The results clearly show the advantage of RSD on congested instances. While Frank-Wolfe required more than one hundred iterations, RSD reached the required tolerance after only seven iterations by optimizing over the convex hull of several previously generated extreme points.

3.4 Comparison with the Direct Nonlinear Solution

The equilibrium problem was also solved directly through its nonlinear Beckmann formulation using AMPL.

The resulting objective value was

$$z_{direct} = 24\,862\,527.44.$$

Table 1 compares the three approaches.

Table 1: Comparison between the three solution approaches.

Method	Iterations	Objective Value	Relative Gap
Direct nonlinear AMPL	–	24,862,527.44	–
Frank-Wolfe	101	24,869,762.38	0.000463
RSD ($\rho = 4$)	7	24,862,823.17	0.000356

The experiment confirms the well-known practical advantage of the RSD method on congested traffic assignment problems. While Frank-Wolfe required 101 iterations to satisfy the stopping criterion, RSD achieved convergence after only 7 iterations.

Both methods produced objective values very close to the direct nonlinear solution. However, RSD obtained a significantly more accurate approximation while requiring substantially fewer iterations.

3.5 Evolution of the Relative Gap and Objective Function

Figure 1 shows the logarithm of the relative gap versus the iteration number for both algorithms. Frank-Wolfe decreases the gap gradually and requires 199 iterations. In contrast, RSD reduces the gap much faster and reaches the stopping tolerance in only 13 iterations.

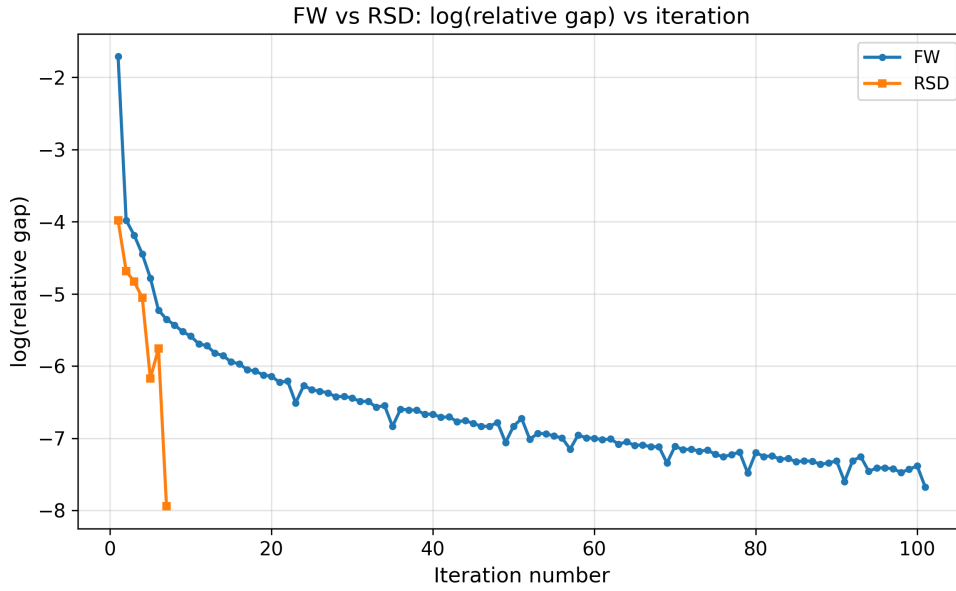


Figure 1: Logarithm of the relative gap versus iteration number.

Figure 2 shows the logarithm of the objective function versus the iteration number. Both methods approach the same final objective value, but RSD does so with significantly fewer iterations.

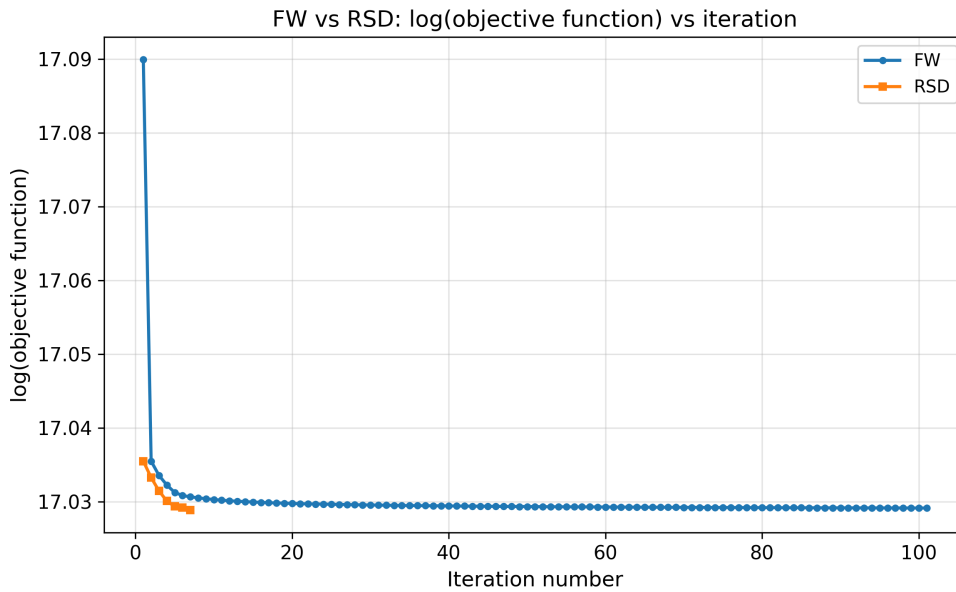


Figure 2: Logarithm of the objective function versus iteration number.

3.6 Iteration Printout

The full iteration outputs are stored in the files `FW_results.txt` and `RSD_results.txt`. These files include, for each iteration, the iteration number, the objective value, the relative gap, the step size or barycentric coordinates, and the number of active vertices in the case of RSD.

For readability, only a summary is included in the main body of the report. The complete printouts should be included together with the AMPL files in the final

delivery. In the following figure we can see an example of the printouts of the .run file and the text files with the results of each algorithm.

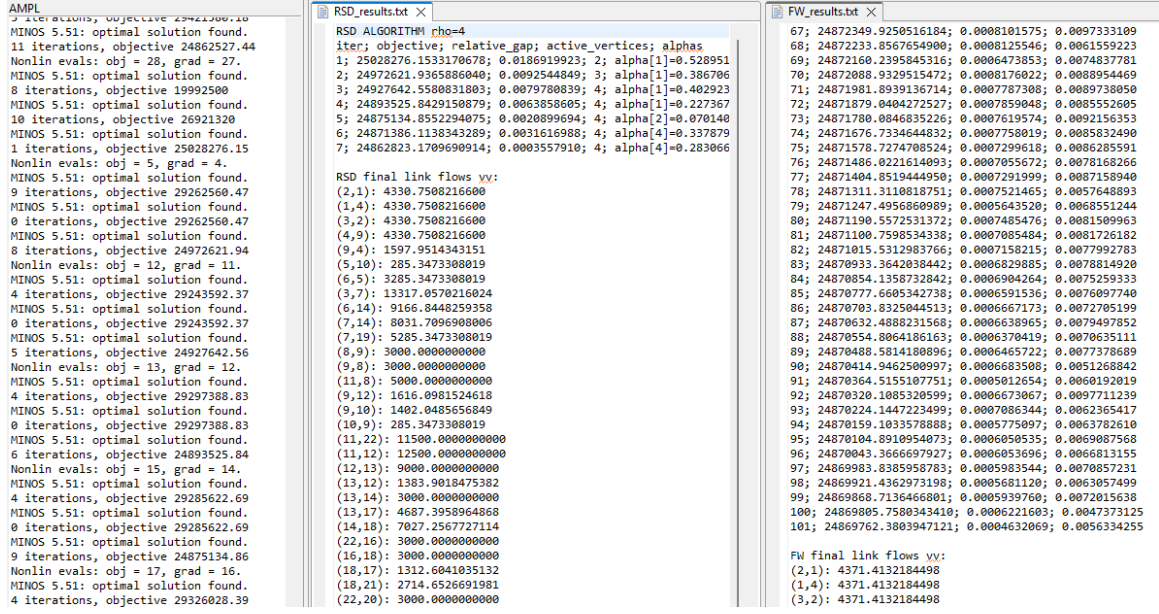


Figure 3: Iteration Printout

4 Wardrop Equilibrium Analysis

The Wardrop equilibrium principle states that all used paths between the same origin and destination must have minimum travel cost. Therefore, if two paths are both used, their costs should be approximately equal. On the other hand, a path with larger travel cost should not be used in the equilibrium solution.

The travel cost of a path P is computed as

$$C(P) = \sum_{(i,j) \in P} s_{ij}(v_{ij}) = \sum_{(i,j) \in P} (c_{ij} + dv_{ij}),$$

where v_{ij} is the final equilibrium flow on arc (i, j) .

The following analysis uses the final RSD solution, since it is the closest one to the direct nonlinear solution.

4.1 Origin-Destination Pair (3, 14)

Two paths from node 3 to node 14 are considered:

$$P_1 : 3 - 7 - 14, \quad P_2 : 3 - 6 - 14.$$

Table 2: Path costs for OD pair (3, 14).

Path	Travel cost	Used in RSD solution?
3 - 7 - 14	628.47	Yes
3 - 6 - 14	628.60	Yes

Both paths are used in the final solution and their travel costs are almost identical. This is consistent with Wardrop's first principle: used paths between the same OD pair should have approximately equal minimum cost.

4.2 Origin-Destination Pair (3, 13)

Two paths from node 3 to node 13 are considered:

$$P_1 : 3 - 6 - 5 - 10 - 13, \quad P_2 : 3 - 7 - 14 - 13.$$

Table 3: Path costs for OD pair (3, 13).

Path	Travel cost	Used in RSD solution?
3 - 6 - 5 - 10 - 13	844.73	Yes
3 - 7 - 14 - 13	845.32	Yes

Again, both paths are used and their costs are very close. The small difference is due to numerical tolerance and the fact that the algorithm stops once the relative gap is below 0.0005, not when exact equality is reached.

4.3 Origin-Destination Pair (3, 8)

Two paths from node 3 to node 8 are considered:

$$P_1 : 3 - 2 - 1 - 4 - 9 - 8, \quad P_2 : 3 - 6 - 5 - 10 - 9 - 8.$$

Table 4: Path costs for OD pair (3, 8).

Path	Travel cost	Used in RSD solution?
3 - 2 - 1 - 4 - 9 - 8	1000.60	Yes
3 - 6 - 5 - 10 - 9 - 8	1001.35	Yes

The two paths have similar costs and both are used. This also supports the interpretation that the final solution is close to a Wardrop equilibrium.

4.4 Origin-Destination Pair (11, 21)

Two paths from node 11 to node 21 are considered:

$$P_1 : 11 - 22 - 20 - 21, \quad P_2 : 11 - 22 - 16 - 18 - 21.$$

Table 5: Path costs for OD pair (11, 21).

Path	Travel cost	Used in RSD solution?
11 - 22 - 20 - 21	845.46	Yes
11 - 22 - 16 - 18 - 21	1019.89	Yes

In this case, the first path is clearly cheaper than the second one. The second path still has positive link flows, but this does not necessarily mean that this exact OD path is used as a route for OD pair (11,21), because the reported solution is link-based and aggregates flows from all OD pairs. Therefore, for this OD pair, the first path is the one that clearly represents a minimum-cost route.

4.5 Origin-Destination Pair (11, 19)

Two paths from node 11 to node 19 are considered:

$$P_1 : 11 - 22 - 23 - 24 - 19, \quad P_2 : 11 - 12 - 13 - 14 - 7 - 19.$$

Table 6: Path costs for OD pair (11, 19).

Path	Travel cost	Used in RSD solution?
11 - 22 - 23 - 24 - 19	1160.96	Yes
11 - 12 - 13 - 14 - 7 - 19	1204.43	No

Here, the first path has lower cost and is used. The second path has a larger cost and contains at least one link with zero flow in the RSD solution. This is consistent with Wardrop's principle: paths with higher cost should not be used when cheaper alternatives are available.

4.6 Discussion of Wardrop Equilibrium

The path analysis is consistent with the Wardrop equilibrium condition. For several origin-destination pairs, two used paths have almost equal costs, which indicates that the flow is split between routes of approximately minimum cost. In cases where one path is clearly more expensive, that path is not used, or at least it is not supported by positive flow on all of its links.

It is important to remark that the AMPL model is link-based rather than path-based. Therefore, checking whether a specific path is used must be interpreted carefully. A path whose links all have positive flow is compatible with being used, but the model does not explicitly store path-flow variables. Even with this limitation, the computed path costs show the expected behaviour of a Wardropian equilibrium.

5 Conclusions

In this assignment, the Frank-Wolfe and Restricted Simplicial Decomposition algorithms were implemented in AMPL for a nonlinear traffic equilibrium problem. The network data were taken from the previous assignments, and congestion was introduced through linear volume-delay functions.

The Frank-Wolfe algorithm converged in 199 iterations, while the RSD algorithm converged in only 13 iterations. This confirms the expected behaviour: RSD requires solving a master problem over a set of active vertices, but it can reduce the number of major iterations substantially.

The comparison with the direct nonlinear AMPL solution shows that both methods approach the same equilibrium objective value. The RSD objective is almost identical to the direct nonlinear objective, while the Frank-Wolfe solution remains slightly farther away due to its slower convergence near the optimum.

Finally, the path cost analysis supports the interpretation of the solution as a Wardrop equilibrium. Used paths tend to have very similar travel costs, while paths with larger costs are not attractive and are not used in the final solution. Overall, the implementation confirms both the practical behaviour of the algorithms and the theoretical properties of user equilibrium problems.